

Course Objectives

- To provide an understanding of basic building blocks of digital circuits
- To enable the understanding of Boolean algebra and logic function optimization
- To enable design of combinational and sequential circuits

Course Outcomes

CO1: Realize a given expression in terms of basic building blocks.

CO2: Minimize a given logic expression.

CO3: Design combinational circuits

CO4: Design Sequential circuits.

CO-PO Mapping

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO														
CO1	3	2												2
CO2	3	2												2
CO3	3	3	2									2		2
CO4	3	3	2									2		2

Syllabus

Unit 1: Introduction to logic circuits - Variables and functions, inversion - Truth tables - Logic gates and Networks - Boolean algebra - Synthesis using gates - Design examples - Optimized implementation of logic functions - Karnaugh map - Strategy for minimization - Minimization of product of sums forms - Incompletely specified functions - Multiple output circuits - Tabular method for minimization.

Unit 2: Combinational circuit building blocks - Number representation and arithmetic circuits: Addition of unsigned numbers - Signed numbers - Fast adders - Multiplexers - Decoders - Encoders - Code converters - Arithmetic comparison circuits.

Unit 3: Sequential circuit building blocks - Basic latch - Gated SR latch - Gated D latch - Master slave and edge triggered - D flip-flops - T flip-flop - JK flip-flop, Registers, Asynchronous Counters, Synchronous Counters, Ring Counter and Johnson Counter, Synchronous sequential circuits - Basic design steps - State assignment problem – Design of Mealy and Moore state models.

Textbook(s)

Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital logic with Verilog Design", Tata McGraw Hill Publishing Company Limited, Special Indian Edition, 2007.

R. D. Sudhakar Samuel, "Logic Design: A Simplified Approach", Sanguine Technical Publishers, Edition 1, 2006

Reference(s)

M Morris Mano and Michael D Ciletti, "Digital Design with Introduction to the Verilog HDL", Pearson Education, Fifth Edition, Fifth Edition, 2015

John F. Wakerly, "Digital Design Principles and Practices", Fourth Edition, Pearson Education, 3rd Ed, 2008.

Donald D Givone, "Digital Principles and Design", Tata McGraw Hill Publishing Company Limited, 2003.

Evaluation Pattern: 60:40

Assessment	Internal	External
Mid Sem Examination	30	
*Continuous Assessment (CAT)	30	
End Semester	2	40

*CAT includes Quizzes, Assignments and Tutorials